

DATABASE DESIGN DOCUMENT

PART-2: ENTITY RELATIONSHIP STANDARDS & MASTER DATA

School ERP

PostgreSQL | Spring Data JPA | Hibernate

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Preamble

This document is Part-2 of the Database Design Document (DDD) v1.0 for the School ERP System, continuing directly from Part-1: Database Foundation. It defines the logical entity relationship, master data, and ownership standards governing all 39 entities already established in Appendix-H, consolidated with the naming, key, and column standards already fixed in DDD Part-1.

This document does not regenerate RGD v2.0, any prior Appendix, any UI/UX Screen Specification, or any prior MDB/DDD Part. It contains no SQL, no CREATE TABLE statements, no entity class definitions, no ER diagram, no UML diagram, no sample data, and no illustrative examples — every standard is stated as a rule, and every matrix presented (Sections 19-20) reuses only the real entity/module names already approved in Appendix-G/H, not invented illustrations.

11. Entity Classification Standards

Fixes the four-way data classification already established in Appendix-G Sections 8-11 and Appendix-H Section 10.4 as a mandatory physical-design attribute of every table.

Classification	Definition	Physical Design Implication
Master Data	Structural reference data changed rarely, read frequently (Class, Section, Subject, AcademicSession, GradingScheme, Route, Vehicle, Department, Designation, FeeHead, Book, Role)	Eligible for caching (DDD Part-1 Section 3; MDB Part-3 Section 26); indexed for read performance over write performance
Transaction Data	High-volume, write-heavy operational records (Application, AttendanceRecord, MarksRecord, Invoice, Payment, BookIssue, PayrollRun, etc.)	Primary target for partitioning (DDD Part-1 Section 3) and the archival policy already fixed in Appendix-H Section 15
Reference Data	Versioned, term-scoped structural data (TimetableEntry)	Retains historical versions rather than being overwritten in place, per Appendix-H Section 4
Configuration Data	Runtime-editable policy values (Configuration entity)	Every write is itself audited (DDD Part-1 Section 7) and treated as a governed change, not routine transactional data

Every table's classification is recorded alongside its Appendix-G/H Entity ID and is never left ambiguous — a table without a stated classification is treated as an incomplete design, not defaulted to Transaction Data.

12. Entity Relationship Standards

Fixes how every one of the 45 relationships already catalogued in Appendix-H Section 5 is documented and physically realized, without altering any relationship's substance.

- Every relationship is documented with a plain-language business name (e.g., "Student enrolls in Class"), exactly as already established for all 45 relationships in Appendix-N Section 5.1 — this naming is carried forward unchanged, not re-derived.
- Every relationship is Non-Identifying by default (the child has its own independent surrogate primary key), consistent with Appendix-H v1.1 Section 3.1 — Identifying relationships are used only within junction tables (Section 14 this document), where the composite key is inherently identifying.
- Relationship direction (parent to child) matches the Foreign Key direction exactly — the table holding the foreign key column is always the child in the relationship documentation, with no exceptions.
- Every relationship's Mandatory/Optional status (Appendix-H Section 5) directly determines whether its Foreign Key column is NOT NULL or nullable at the physical level — this is a direct, one-to-one mapping, not a separate design decision.

13. Relationship Cardinality Rules

Fixes the cardinality notation already introduced in Appendix-N Section 5.1 as the mandatory documentation standard for every relationship in the physical design.

Notation	Meaning	Physical Design Rule
(1)	Exactly one — mandatory, singular	Foreign Key is NOT NULL; at most one matching parent row is ever valid
(0..1)	Zero or one — optional, singular	Foreign Key is nullable; at most one matching row when present
(1..*)	One or many — mandatory, plural	Child side of a One-to-Many; at least one child row expected once the parent exists
(0..*)	Zero or many — optional, plural	Child side of a One-to-Many where zero child rows is a valid, expected state

Many-to-Many relationships are always expressed as (1..*) or (0..*) on both sides simultaneously, resolved through a junction table (Section 14) — a Many-to-Many relationship is never approximated as a single Foreign Key on either participating table.

14. Junction Table Standards

Fixes the physical design pattern for the 3 junction tables already fully specified in Appendix-H Section 6 — RolePermission, ClassSubjectMap, StudentGuardianLink — as the template for any future Many-to-Many relationship.

- A junction table's primary key is the composite of its two Foreign Key columns, per DDD Part-1 Section 9's stated exception to the single-column-surrogate rule — no junction table introduces a separate surrogate ID column in addition to the composite key.
- Every junction table still carries the Common Columns baseline (DDD Part-1 Section 7: version, audit columns, soft-delete) — a junction table is a full table subject to the same governance as any other, not a lightweight exception.
- A junction table's two Foreign Keys both enforce RESTRICT on delete, consistent with the no-Cascade-Delete rule (DDD Part-1 Section 9) — removing a Role does not silently cascade-remove its RolePermission rows; the association must be explicitly cleared first.
- Reverse-lookup indexing (e.g., "which roles have X permission" for RolePermission) is created on the second column of the composite key in addition to the composite index itself, per the specific indexing note already given per junction table in Appendix-H Section 6.

15. Master Data Standards

Fixes governance rules for the Master Data classification already defined in Section 11, extending the Data Ownership Matrix already established in Appendix-G Section 13.

- Every Master Data table has exactly one designated Business Owner role (per Appendix-G Field 5, e.g., Academic Head for Class/Section/Subject, Finance Team for FeeHead) authorized to create and update records — Master Data is never editable by an unrestricted set of roles.
- Master Data changes are logged with the same audit rigor as transactional changes (DDD Part-1 Section 7, MDB Part-2 Section 20), since a Master Data error (e.g., an incorrect FeeStructure amount) has broad downstream impact across many transactional records.
- Master Data is never hard-deleted, and a Master Data record already referenced by any transactional row cannot be soft-deleted either without first reassigning or closing out the referencing rows — this is a stricter deletion rule than the general soft-delete baseline (DDD Part-1 Section 7), reflecting Master Data's structural role.
- Historical Master Data versions (e.g., a superseded FeeStructure or GradingScheme) are retained, not overwritten, so that historical transactional records remain fully interpretable — consistent with Appendix-H's treatment of FeeStructure and GradingScheme as versioned per academic session.

16. Lookup Data Standards

Reuses the Lookup Table gap analysis already completed in Appendix-N Section 7 without re-deriving it — this section fixes the physical-design consequence of that analysis.

Lookup Concept	Physical Design Status
Gender, Religion, Blood Group	Not Defined in Approved Scope — no corresponding attribute or table exists; not implemented
Category, Fee Status, Attendance Status, Result Status, Transport Status, Notification Status, User Status	Implemented as an enumerated (constrained-value) column on the owning table, not a separate lookup table — matches Appendix-N Section 7's finding exactly
Role, Academic Session, Class, Section, Subject, Department, Designation, Fee Head, Grading Scheme	Implemented as a full Master Data table (Section 15), each with its own Entity ID
Permission	Resolved via the RolePermission junction table (Section 14); no standalone Permission master table exists, consistent with Appendix-G's Missing Data Analysis

This table is not re-derived here — it is the same finding already reached in Appendix-N Section 7, restated at the physical-design level so a Database Administrator does not need to cross-reference back to that document to know which lookups are real tables versus enumerated columns versus genuinely absent.

17. Cross-Module Relationship Standards

Reuses the confirmed cross-module relationships already established in Appendix-I Section 9, fixing how they are physically realized without altering which relationships exist.

- Every cross-module Foreign Key (e.g., Invoice.student_id reaching into the Student Information module from Fees) is a standard, database-enforced Foreign Key exactly as documented in Appendix-H Section 5 — a cross-module reference is not treated differently from a within-module reference at the physical level.
- Cross-module effects that are not expressed as a direct Foreign Key (Library fine posting to Fees, Transport fee linkage, Communication notification triggers) remain application-layer event-driven, per MDB Part-1 Section 8 — no additional database trigger or stored procedure is introduced to implement these effects at the database tier.
- The Student and Employee entities remain the highest-fan-out tables in the physical schema, exactly as already identified in Appendix-D's Critical Dependency List and Appendix-I Section 4.3 — any schema change to either table requires the broadest cross-module impact review of any table in the system.

18. Entity Ownership Standards

Fixes the Business Owner / Technical Owner distinction already established in Appendix-G Section 13 as a mandatory, documented attribute of every table.

- Every table's Business Owner is the role already assigned in Appendix-G Field 5 — this role is accountable for the correctness of the data, not for its physical storage.
- Every table's Technical Owner is IT Admin uniformly, exactly as already fixed in Appendix-G Field 18 — no table has a module-specific Technical Owner, since physical database administration is a centralized, not module-distributed, responsibility.
- A schema change request (DDD Part-1 Section 10, MDB Part-3 Section 28) requires sign-off from the affected table's Business Owner in addition to the standard Code Review Checklist (MDB Part-3 Section 30), ensuring business-side awareness of any structural change to data they are accountable for.

19. Entity Dependency Matrix

Module-level summary of the entity dependency chain already fully detailed at the entity level in Appendix-D and Appendix-H Section 5 — presented here as a table, not a diagram, per this document's no-diagram constraint.

Module	Depends On	Depended On By
Admission	Academic Master (Class)	Student Information
Student Information	Admission	Attendance, Examination, Fees, Library, Transport, Reports, Communication
Attendance	Student Information, Timetable	Examination (eligibility), HR & Payroll (staff variant), Reports
Examination	Student Information, Attendance, Academic Master	Reports, Communication
Fees	Student Information, Academic Master	Library (fine posting), Transport (fee linkage), Reports
Library	Student Information, HR & Payroll (staff borrowers)	Fees (fine posting)
Transport	Student Information	Fees (fee linkage)
HR & Payroll	Academic Master (indirect, via Attendance)	Attendance (staff), Library (staff borrowers), Reports
Communication	Admission, Attendance, Fees, Examination (event triggers)	None (terminal consumer)
Reports	Attendance, Examination, Fees, HR & Payroll, Library, Transport (read-only)	None (terminal consumer)
Administration	None (foundational)	All other modules (shared services)

20. Entity Traceability Matrix

Confirms that every module's entities remain traceable to their governing Functional Requirements and Business Rules, reusing the ID mappings already fixed in Appendix-E, Appendix-C v1.1, and Appendix-H Section 4 without restating full entity-level detail.

Module	Governing FR Range	Governing BR Range	Entity ID Prefix (Appendix-H)
Admission	FR-01 to FR-05c	BR-ADM-001 to BR-ADM-008	ENT-ADM
Student Information	FR-06 to FR-09	BR-SIS-001 to BR-SIS-006	ENT-SIS
Attendance	FR-10 to FR-13	BR-ATT-001 to BR-ATT-007	ENT-ATT
Examination	FR-17 to FR-21	BR-EXM-001 to BR-EXM-007	ENT-EXM
Fees	FR-22 to FR-26b	BR-FEE-001 to BR-FEE-008	ENT-FEE
Library	FR-27 to FR-29	BR-LIB-001 to BR-LIB-006	ENT-LIB
Transport	FR-30 to FR-32	BR-TRN-001 to BR-TRN-006	ENT-TRN
HR & Payroll	FR-33 to FR-36	BR-HR-001 to BR-HR-007	ENT-HR
Communication	FR-37 to FR-39	BR-COM-001 to BR-COM-005	ENT-COM
Reports	FR-40 to FR-42	BR-RPT-001 to BR-RPT-005	N/A (no entities of its own)
Administration	N/A (cross-cutting)	Shared rules (e.g., BR-HR-002 access revocation)	ENT-SYS

Every table created against this DDD is expected to carry its Appendix-G/H Entity ID in its migration documentation (MDB Part-3 Section 28), preserving the unbroken traceability chain from RGD v2.0 through the physical schema itself.